

HD189733b

R-1127

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HD189733_220725 · created 2026-07-30T22:33:43+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459785.9433 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.128 +/- 0.027
Transit depth (Rp/Rs)^2	1.63 +/- 0.69 [%]
Orbital Inclination (inc)	87.41 +/- 1.62
Airmass coefficient 1 (a1)	1.002 +/- 0.022
Airmass coefficient 2 (a2)	0.0 +/- 0.012
Scatter in the residuals of the lightcurve fit is	2.19 %
Best Comparison Star	#2 - [404, 293]
Optimal Aperture	4.39
Optimal Annulus	6.75
Transit Duration (day)	0.0795 +/- 0.0098

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.128 $\pm$ 0.027 vs 0.15534 (deviation 0.78 σ)
Duration fit vs published	0.0795 d vs 0.0754 d (deviation 0.32 σ)
Mid-time O-C	4.1 min
Depth SNR	3.7
Residual scatter	2.19 %

Light curve

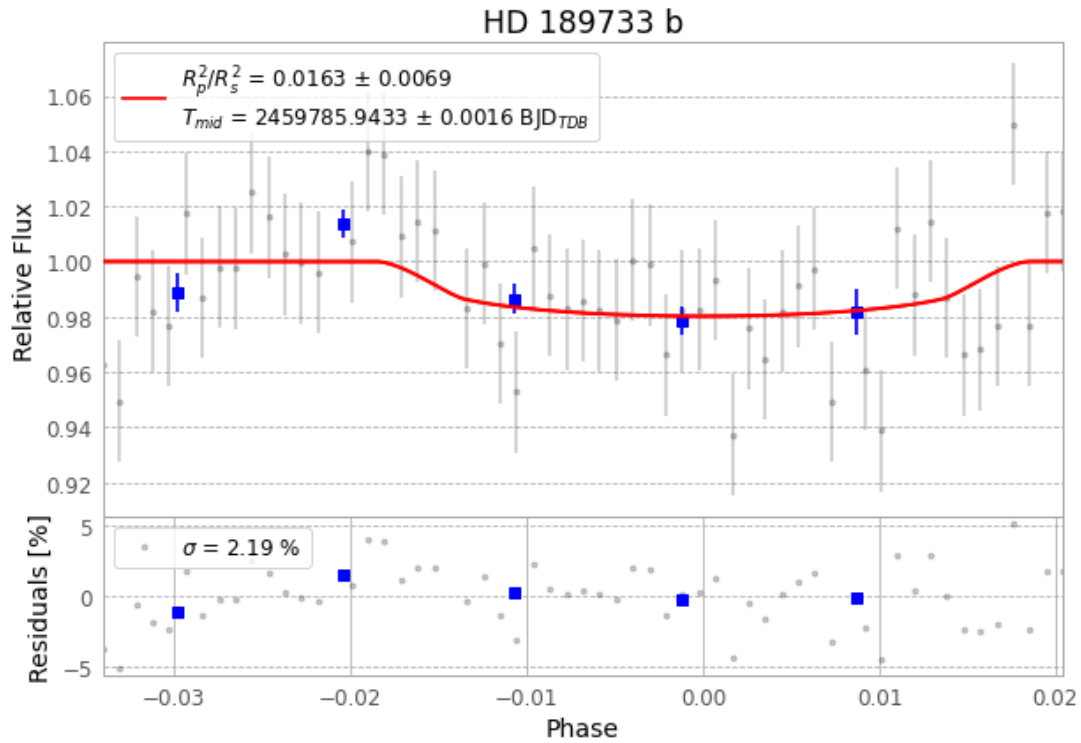


Figure 1 — FinalLightCurve_HD 189733 b_2022-07-25.png (SHA-256 70e74bdf9da22dff...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [304, 288], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e9f0d87c55e177a6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

59 FITS frames (39 MB), HD189733220725084213.FITS → HD189733220725113612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HD189733-220725.log (974faafc1788...), FinalLightCurve_HD 189733 b_2022-07-25.png (70e74bdf9da2...), FinalLightCurve_HD 189733 b_2022-07-25.csv (e57abdeacc83...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 22:33Z.